

Qn	Suggested Solutions
4(a)	A polarised wave has its vibrations/oscillations occur in a single direction in a plane / restricted to one plane (axis) and perpendicular to the direction of transfer of energy/propagation.
(b)	Applying Malus' Law and assuming that the intensity of the polarised light after polaroid Q is I_0 , $I = I_0 \cos^2 \theta$ $= I_0 \cos^2 (30^\circ)$ $= 0.75 I_0$
	Since $I \propto A^2$ $\frac{I}{I_0} = \left(\frac{A}{A_0} \right)^2$ $\cos(30^\circ) = \frac{A}{A_0}$ $A = 0.866 A_0$ $\approx 0.87 A_0$
(c)	It is the spreading of the wave into its geometrical shadow when it is incident on an aperture/opening or edge of an obstacle.
(d)(i)	$d \sin \theta = n\lambda$ $(3.4 \times 10^{-6}) \sin \left(\frac{16^\circ}{2} \right) = \lambda$
	$\lambda = 4.73 \times 10^{-7} \text{ m}$ $\approx 4.7 \times 10^{-7} \text{ m}$
(ii)	Blue/Indigo
(iii)	For an emerging beam to be observed on the screen, $\theta \leq 90^\circ$ and hence $\sin \theta \leq 1$.
	Since $\sin \theta = \frac{n\lambda}{d}$, $\frac{n\lambda}{d} \leq 1$ $n \leq \frac{d}{\lambda}$ $n \leq \frac{3.4 \times 10^{-6}}{4.73 \times 10^{-7}}$ $n \leq 7.19$
	no. of emerging beams = $7 + 7 + 1 = 15$
(iv)	Without the polaroids, the incident light intensity on the diffraction increases, so the diffraction maxima become brighter.
	The separation between maxima depends only on the wavelength of light and the line spacing of the grating, which are unchanged, so it remains the same.

Anglo-Chinese Junior College

H2 (9749) Physics

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